

Prediction of Protein–Ligand Binding Pose and Affinity Using the gREST+FEP Method

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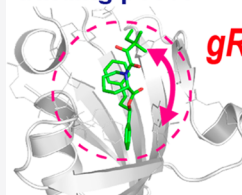
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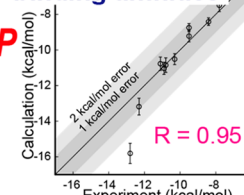
Supporting Information

ABSTRACT: The accurate prediction of protein–ligand binding affinity is a central challenge in computational chemistry and *in-silico* drug discovery. The free energy perturbation (FEP) method based on molecular dynamics (MD) simulation provides reasonably accurate results only if a reliable structure is available via high-resolution X-ray crystallography. To overcome the limitation, we propose a sequential prediction protocol using generalized replica exchange with solute tempering (gREST) and FEP. At first, ligand binding poses are predicted using gREST, which weakens protein–ligand interactions at high temperatures to sample multiple binding poses. To avoid ligand dissociation at high temperatures, a flat-bottom restraint potential centered on the binding site is applied in the simulation. The binding affinity of the most reliable pose is then calculated using FEP. The protocol is applied to the bindings of ten ligands to FK506 binding proteins (FKBP), showing the excellent agreement between the calculated and experimental binding affinities. The present protocol, which is referred to as the gREST+FEP method, would help to predict the binding affinities without high-resolution structural information on the ligand-bound state.

Prediction of binding poses



Prediction of binding affinities



1. INTRODUCTION

The accurate prediction of protein–ligand binding affinity is one of the key challenges in computational chemistry and *in-silico* drug design because of its potential to reduce the cost and time for drug development. X-ray crystal structures of target protein–ligand complexes give structural information on the binding sites and poses, which is essential to predict the binding affinities. Even when the binding site is known, there may exist multiple binding poses in the site. Since the binding affinity depends severely on the binding pose, the determination of reliable poses is a prerequisite for high-precision drug design.^{1–3} To date, a variety of computational methods have been proposed for pose prediction.

Protein–ligand docking methods predict multiple binding poses and calculate their affinities.^{4–10} Since the docking methods are fast and computationally efficient, they are widely used to virtually screen potential drug candidates. However, protein flexibility is not sufficiently considered in most of the docking simulations, while ligand bindings often couple with protein conformational changes. Many different types of scoring functions have been developed, although they often simplify the protein–ligand interactions based on the shape complementarity and treat solvation effects implicitly, missing physical components including entropy contributions from solvent molecules and proteins. Docking methods are less accurate in ranking the predicted poses and sometimes overlook plausible binding poses.^{11–14}

Molecular dynamics (MD) simulations provide protein structure and dynamics at atomic resolution. The simulations

explicitly include all degrees of freedom relevant to protein–ligand bindings including solvent motions. When a high-resolution X-ray crystal structure is available, binding affinities of small compounds can be accurately predicted using the free energy perturbation (FEP) or thermodynamic integration method.^{15–17} The most stable binding pose can be predicted precisely from long MD trajectories, in principle, even if high resolution structures of the complex are unavailable. However, the time scale of the binding pose interconversion is extremely slow, so that only a few binding poses can be sampled by microsecond-scale MD simulations.¹⁸ The ligand is often trapped at metastable poses, missing the most stable binding pose in the simulations.

To sample multiple binding poses sufficiently, enhanced sampling methods have been combined with MD simulations. For example, the combination of alchemical techniques and replica-exchange methods have been proposed.^{2,19,20} In alchemical simulations, the protein–ligand interactions are scaled using an alchemical parameter λ . To enhance the sampling efficiency, the alchemical simulations at different λ values are coupled using the Hamiltonian replica exchange

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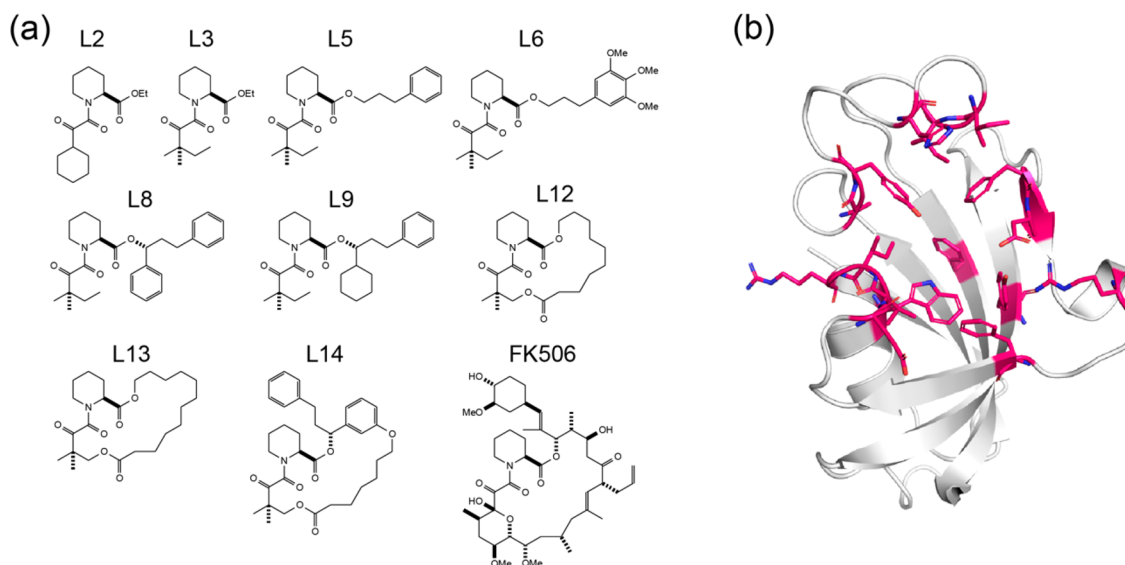


Figure 1. (a) Chemical structures of ten ligands for FKBP. (b) Structure of apo FKBP. The residues selected as the solute region of gREST are shown in magenta stick.

method.^{21,22} This method effectively weakens the protein–ligand interactions and enables the ligand to reorient in the site, accelerating the sampling of multiple poses. An alchemical MD simulation combined with a nonequilibrium Monte Carlo approach has also been proposed.^{18,23} In the method, the ligand orientation, which is sampled through the alchemical perturbation, is accepted or rejected at a nonequilibrium Monte Carlo step. A nonequilibrium process enables large movements of the ligand with high acceptance.

Here, we make use of generalized replica exchange with solute tempering (gREST)²⁴ to predict the binding poses. gREST is a generalization of REST²⁵ and provides a flexible framework for solute definition. The solute can be defined as either a part of a molecule and/or a part of the potential energy terms. Similar to the alchemical simulations, gREST weakens the protein–ligand interactions at high temperature replicas, accelerating the ligand reorientation or rotation in the binding site. The method, with a proper solute definition, is also able to make the protein binding site residues flexible and water diffusion near the binding pocket faster.^{26,27}

We apply gREST to the bindings of ten ligands to FK506 Binding Protein (FKBP).²⁸ For this system, the experimental binding affinities are available and there exist cocrystal structures for four ligands (L8, L9, L13, and FK506).^{28,29} To show the reliability of our method, however, we predict binding poses and affinities without using the prior knowledge. To avoid ligand dissociations at high temperatures, a flat-bottom (FB) restraint potential centered on the binding site is applied.²⁷ gREST with the FB potential provides the multiple binding poses for each complex and successfully reproduces the crystal structures. Their binding affinities are then evaluated using the FEP method implemented in version 1.5.0 of the GENESIS program package (<https://www.r-ccs.riken.jp/labs/cbrt/>).^{30,31} The calculated binding affinities of the predicted poses are in excellent agreement with the experimental results. The method is general and requires only a small number of replicas. The presented protocol, referred to as gREST+FEP, would help to predict the binding affinities of ligands without high-resolution structural information on the bound state.

2. METHODS

2.1. Modeling of Ligands and a Target Protein. The Gaussian 09 program³² was used to optimize the geometries of ten ligands (Figure 1a) and to calculate the electrostatic potentials at the Hartree–Fock/6-31G* level. The atomic partial charges were obtained by the restrained electrostatic potential approach.³³ The remaining parameters for the ligands were determined by the general Amber force field³⁴ (GAFF2) using the antechamber module in Amber Tools 16.³⁵

Initial structures of the FKBP–ligand complexes were obtained by docking simulations using Autodock Vina.⁹ We assumed that the canonical binding site is known *a priori*. The atomic model of FKBP was constructed from the apo crystal structure (PDBID 2PPN;³⁶ Figure 1b). Missing side-chain atoms were added using Leap module in Amber Tools 16. We used HIE for all histidine residues, because we assumed their tautomer (HIE/HID) states does not largely affect interactions between the residues and the ligand. Fifteen residues around the binding sites of the FKBP (Y26, F36, D37, R42, F46, E54, V55, I56, R57, W59, Y82, H87, I90, P92, and F99) were treated as flexible during the docking, while the other residues were kept rigid. The grid size as dimensions of 40 Å in *x*, *y*, and *z*-dimensions was examined and the binding site was located at the grid center. For each ligand, the docked structure with the highest affinity in Autodock Vina was employed as the initial pose for subsequent gREST simulations. Hereafter, the binding poses obtained by Autodock Vina are referred to as “docking poses”.

After obtaining the docking poses, each FKBP–ligand complex was solvated within a rectangular box containing approximately 7800 water molecules. Na⁺ and Cl[−] ions were added to neutralize the system. Each complex system includes roughly 25 000 atoms. The Amber ff14SB force field³⁷ was used for protein and ions, and the transferable intermolecular potential with three points (TIP3P) model³⁸ was used for water molecules. Molecular graphics images were generated using PyMOL.³⁹

2.2. gREST Simulation with a Flat-Bottom Restraint Potential. The details of gREST are described in our earlier studies.^{24,26} In this study, we defined the solute region as the

Lennard-Jones and electrostatic energy terms of a target ligand as well as the binding site residues in FKBP (Y26, F36, D37, R42, F46, E54, V55, I56, R57, W59, Y82, H87, I90, P92, and F99) (Figure 1b). Twelve replicas were used to enhance reorientation and/or rotation of the ligand. Solute temperatures for the 12 replicas are set to 300.0, 324.5, 355.7, 389.5, 431.2, 476.8, 529.2, 593.6, 670.5, 763.7, 871.7, and 999.8 K, while the solvent temperature is kept at 300 K. These parameters were determined to achieve the exchange probability of 0.25. We performed the gREST simulations of ten FKBP–ligand systems using the MD software GENESIS.^{30,31} Each simulation was started from the docking pose. Each system was first equilibrated in a 100 ps NPT (isobaric–isothermal) ensemble with positional restraints, maintaining the temperature and the pressure at 300 K and 1 atm, respectively. After NPT equilibration, a 100 ps NVT (isochoric–isothermal) equilibration at 300 K was performed without any restraints. Each replica in gREST was simulated for 100 ns (1.2 μ s in total). Snapshots of the system at each solute temperature were saved every 1 ps for analysis. Exchanges between neighboring replicas were attempted every 1000 steps (2 ps) and were accepted or rejected according to the Metropolis criterion. After sorting the trajectories by solute temperatures, the trajectory at 300 K (from 50 to 100 ns) was used for analysis. All the gREST simulations were performed in the NVT ensemble.

Ligands can dissociate from the protein at high solute temperatures in gREST simulations. To avoid the ligand dissociation, an FB potential centered on the binding site was applied to each replica in gREST:²⁷

$$U(R) = \begin{cases} K(R - R_0)^2, & R > R_0 \\ 0, & R \leq R_0 \end{cases} \quad (1)$$

where U is the FB restraint potential and R is the distance between centers of mass (COMs) of a ligand and the binding site residues (Y26, F36, D37, R42, F46, E54, V55, I56, R57, W59, Y82, H87, I90, P92, and F99). The midpoint value and force constant were set to $R_0 = 10$ Å and $K = 1$ kcal/mol·Å², respectively. The FB potential thus confines the ligand within the radius of 10 Å, while allowing the ligand to freely rotate in the binding site.

2.3. Pose Determination. We used the distance root-mean-square displacement (dRMS)⁴⁰ to determine a plausible binding pose. dRMS between two structures X^{ref} and X^{target} (docking pose and snapshots; see below) of the protein–ligand complex is defined as

$$\text{dRMS}(X^{\text{ref}}, X^{\text{target}}) = \sqrt{\frac{1}{N_{\text{contacts}}} \sum_{i \neq j \in \text{contacts}} [d_{ij}^{\text{ref}} - d_{ij}^{\text{target}}]^2} \quad (2)$$

where d_{ij} is the distance between protein atom i and ligand atom j in X^{ref} or X^{target} . The difference between d_{ij}^{ref} and d_{ij}^{target} is summed over all atom pairs in a specified contact list. N_{contacts} is the number of the atom pairs. Here we included all pairs between heavy atoms of the protein and the ligand in the contact list. dRMS measures the structural change from the reference to the target structures.⁴¹ Since the matrix $[d_{ij}^{\text{ref}} - d_{ij}^{\text{target}}]$ sensitively reflects the geometrical change of the ligand relative to the protein, dRMS can be used to classify the binding poses. For each protein–ligand complex, we treated

the docking pose as the reference structure (X^{ref}) and calculated dRMS of the complex structure in each snapshot (X^{target}) from the gREST trajectory.

The sampled dRMS values are divided into histogram bins, and the number of samples in each bin is counted. The probability of dRMS, $p(\text{dRMS})$, is defined as the number of samples in the corresponding bin divided by the total number of samples. The potential of mean force (PMF) of dRMS is calculated as follows:

$$\text{PMF}(\text{dRMS}) = -k_B T \ln p(\text{dRMS}) \quad (3)$$

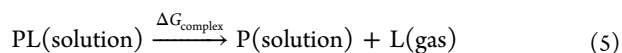
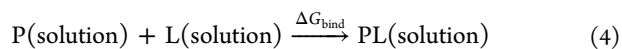
where k_B and T is the Boltzmann constant and the absolute temperature, respectively.

The PMFs of dRMS show not only the global minimum but also several local minima. We selected the binding poses corresponding to the local minima whose PMFs are within 1 kcal/mol of the global minimum. To obtain the representative structure corresponding to each minimum, we calculated the average structure of the snapshots within the minimum, and then calculated the root-mean square deviation (RMSD) between each snapshot and the average structure. Finally, the structure with the minimum RMSD was chosen as the representative binding pose of the minimum.

As with the dRMS analysis, we also calculated the PMFs of RMSD between the gREST trajectory and the docking pose and selected the binding pose corresponding to the global minimum of the PMF of RMSD. In this study, the RMSD analysis considers a ligand and the overall structure of a protein, which means that RMSD depends on not only side-chain contacts with the ligand but also the conformational changes of the protein and ligand.

To compare the predicted pose with the cocrystal structure, we calculated RMSDs of only ligands. We first fitted the α atoms of FKBP (i.e., only the protein part) of the predicted pose to those of the cocrystal structure, and we subsequently calculated RMSD of heavy atoms of ligands between the predicted pose and the crystal structure. This RMSD is referred to as ligand's RMSD to differentiate the RMSD analysis described above.

2.4. Calculation of Binding Affinities. We calculated the binding affinities for the predicted binding poses using the alchemical FEP method implemented in version 1.5.0 of GENESIS.³¹ We employed the MP-CAFE method (massively parallel computation of absolute binding free energy with well-equilibrated states),^{17,42} which is one of the double annihilation methods (DAMs)⁴³ for FEP. In the DAM, the free-energy change upon binding ΔG_{bind} is decomposed into two parts:



where P, L, and PL stand for the protein (FKBP), the ligand, and the protein–ligand complex, respectively. $\Delta G_{\text{complex}}$ and ΔG_{ligand} correspond to the free energy changes when L or PL and L are transferred from solution to gas phase, respectively. ΔG_{bind} can be calculated as

$$\Delta G_{\text{bind}} = \Delta G_{\text{ligand}} - \Delta G_{\text{complex}} \quad (7)$$

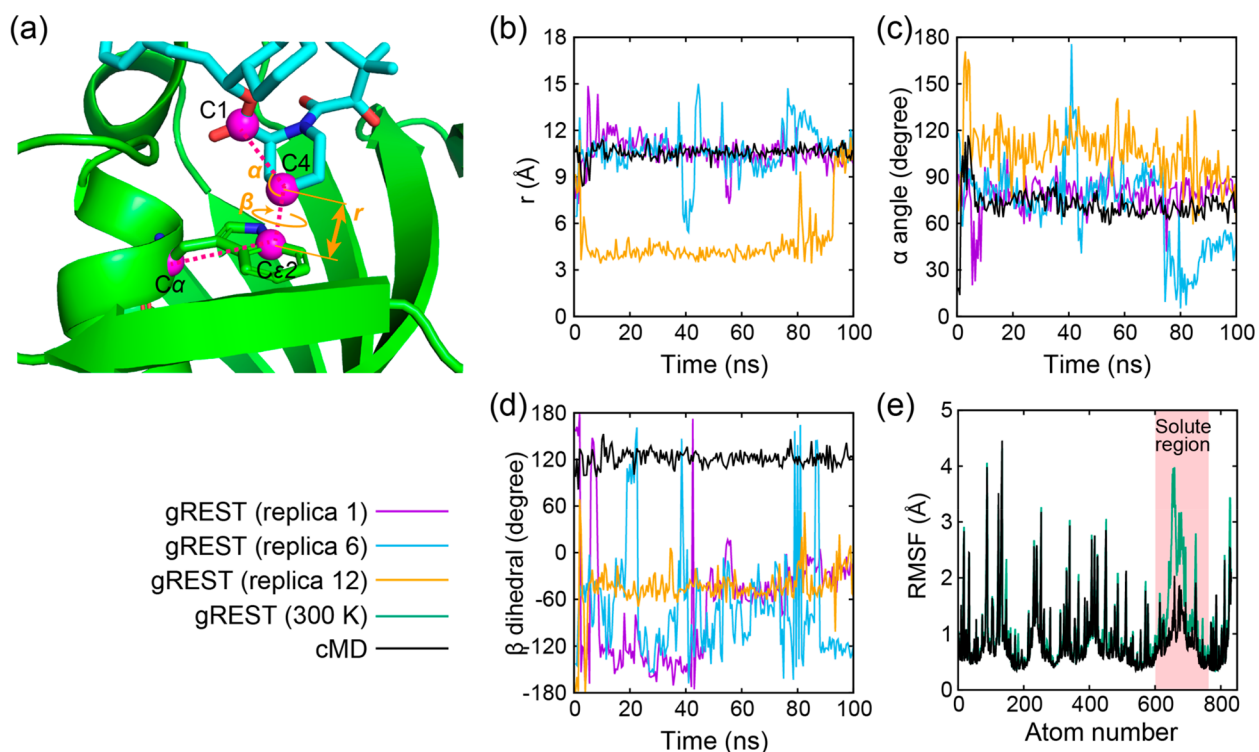


Figure 2. (a) Reaction coordinates for ligand's orientation. FKBP and L9 are shown in green and cyan, respectively. Trp59 of FKBP and L9 are shown in stick representation. (b–d) Time series of r , α angle, and β dihedral, respectively. (e) RMSF of heavy atoms of FKBP. The solute region is defined in Figure 1b. Purple, cyan, and orange lines represent time series of replicas 1, 6, and 12 of gREST with the FB potential, respectively, while the black line represents that of conventional MD. The green line represents the RMSF of gREST with the FB potential at 300 K for both solvent and solute.

In practice, ΔG_{ligand} and $\Delta G_{\text{complex}}$ are calculated by decoupling the interactions between L of PL or L and surroundings using the coupling parameters. In the present study, the electrostatic and Lennard-Jones (LJ) interactions are decoupled by coupling parameters λ^{el} (1.0, 0.9, 0.75, 0.6, 0.45, 0.35, 0.275, 0.2, 0.125, 0.05 and 0.0) and λ^{LJ} (1.0, 0.9, 0.8, 0.7, 0.6, 0.525, 0.45, 0.4, 0.35, 0.325, 0.3, 0.275, 0.25, 0.225, 0.2, 0.175, 0.15, 0.125, 0.1, 0.05, and 0.0), respectively. We applied the same soft-core potential functions as used in ref 26.

The double annihilation method, including MP-CAFEE, requires no restraint potentials in principle. However, as the λ values become smaller, the interactions between a ligand and a protein becomes weaker. Finally, the ligand unbinds from the protein and move freely in the simulation box. In the method, ΔG_{bind} depends thus on the volume of the simulation system (i.e., the volume available to the translational displacement of the free ligand). To remove the system-volume effect, the correction for the standard state is required. The binding affinity ΔG_{bind}^0 was obtained by applying the correction for the standard state:^{15,44}

$$\Delta G_{\text{bind}}^0 = \Delta G_{\text{bind}} - k_{\text{B}}T \ln \left(\frac{V}{V_0} \right) \quad (8)$$

where V is the volume of the simulation system and V_0 is the standard volume. For a standard state of 1 mol/L, V_0 becomes 1,660 Å³. In this study, the box size of the system was approximately 63 Å × 63 Å × 63 Å.

For the calculation of $\Delta G_{\text{complex}}$, the system was equilibrated for 20 ns in the NPT ensemble. Final configurations were used for subsequent FEP calculations. We conducted 5 independent

FEP calculations (NPT ensemble) with different random seeds. In each calculation, the simulation was run for 10 ns per each λ , and trajectories from 5 to 10 ns were used for analysis. Finally, the free-energy differences were estimated using Bennett's acceptance ratio (BAR) method.⁴⁵ For FK506, 30 ns FEP simulations were performed and 20–30 ns trajectories were used for analysis, because the convergence of the free-energy calculation was slow compared to other ligands. The convergence of the free-energy calculation for each system is shown in Figure S1.

For the calculations of ΔG_{ligand} , the same procedure as used for FKBP–ligand complexes was used but with shorter simulation length. For each λ , 5 ns FEP simulations were performed and trajectories from 3 to 5 ns were used for the free-energy calculation.

2.5. Conventional MD. We performed a conventional MD (cMD) simulation, with the FB potential, of the FKBP-L9 system for comparing its sampling efficiency with gREST. The docking pose was employed as the initial structure. The cMD simulation in the NVT ensemble was run for 100 ns. The root-mean-square fluctuation (RMSF) of heavy atoms of FKBP was calculated using the trajectory from 50 to 100 ns.

3. RESULTS

3.1. gREST Enhances Ligand Reorientation. We first examine the sampling efficiency of gREST with the FB potential in comparison with cMD simulation of the FKBP-L9 system. To measure the ligand reorientation and/or rotation, three reaction coordinates are defined: C ϵ 2–C4 distance (r), the angle of C ϵ 2–C4–C1 (α), and the dihedral of C α –C ϵ 2–C4–C1 (β) (Figure 2a). Figure 2b–d show time courses of

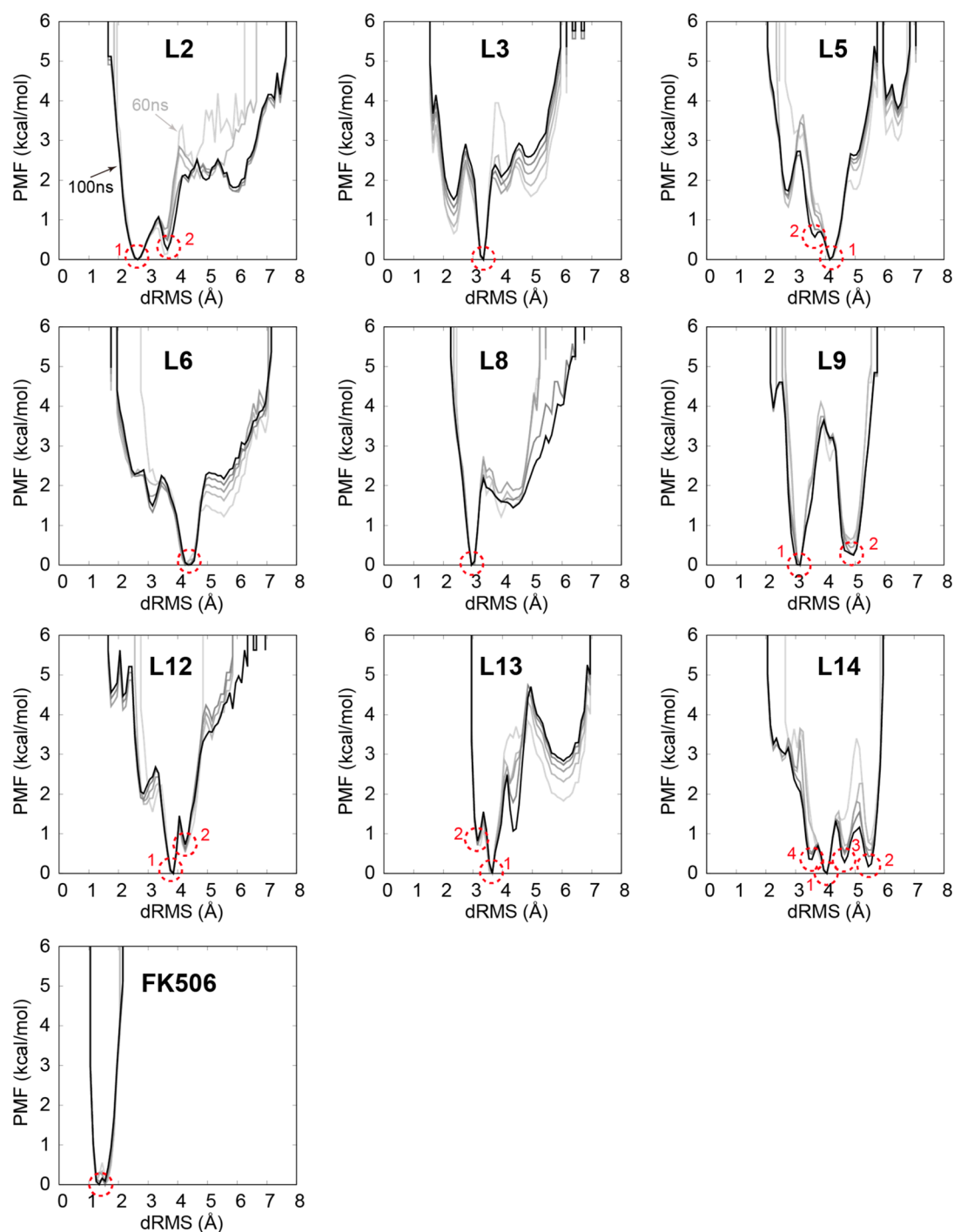


Figure 3. PMFs of dRMS for each FKBP–ligand complex at 300 K. The red dashed circle denotes the basin which is used to determine the representative binding pose (see Figure 4). Gray lines are obtained using trajectories from 50 ns to 60, 70, 80, and 90 ns, while the black line is obtained using trajectories from 50 to 100 ns per each replica.

these reaction coordinates of replicas 1, 6, and 12 of gREST and those of cMD. The reaction coordinates hardly change in the 100 ns cMD simulation, while in the replicas of gREST, the coordinates fluctuate largely. These results imply that the ligand rotational motions in the binding site is significantly enhanced compared to the cMD.

In gREST, the binding-site residues of FKBP were also treated as the solute such that not only the ligand motion but also the flexibility of the protein residues in the binding site are enhanced. The fluctuations of these residues are indeed larger

in gREST at 300 K for both solvent and solute than in cMD (Figure 2e). The increased fluctuations of the binding sites come from the large movement of the binding-site residues in FKBP at high solute temperatures and affects the protein binding pose predictions. Even if the residues and ligand are trapped in metastable states (or metastable binding poses), the enhanced motion of the residues at high solute temperatures allows them to escape from the state, leading to the efficient pose sampling in gREST.

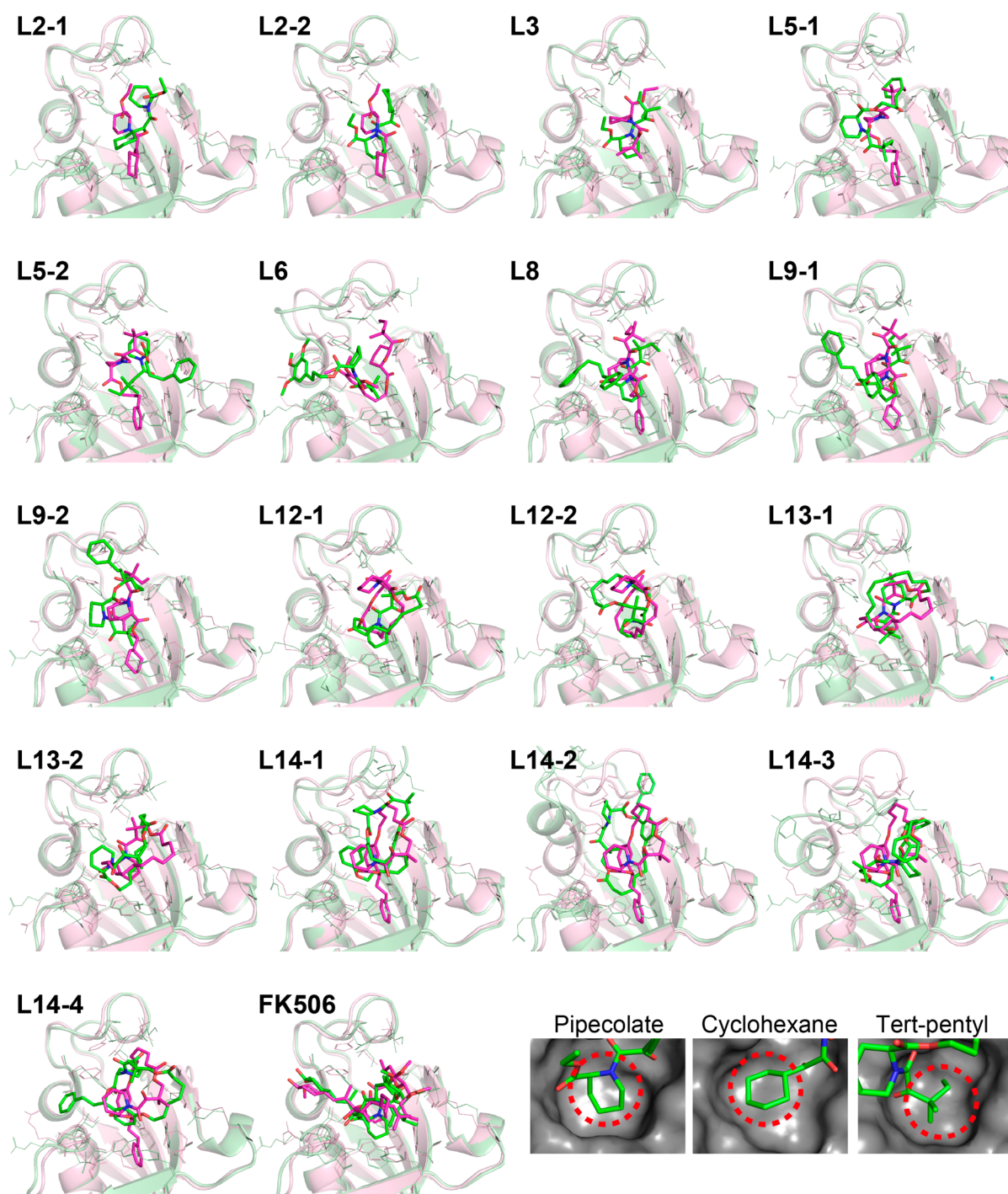


Figure 4. Binding poses for each FKBP–ligand complex. The poses determined by gREST simulations and by Autodock Vina are shown in green and magenta, respectively. The number following the ligand name represents the order of the basin of the PMF as a function of dRMS. (e.g., L2-1 and L2-2 correspond to the most stable and second most stable basins, respectively.) Key chemical groups (pipecolate, cyclohexane, and tert-pentyl) facing to the hydrophobic pocket are magnified on the bottom-right corner.

3.2. gREST Samples Multiple Binding Poses. We predicted the ligand-binding poses in ten FKBP–ligand complexes using gREST with the FB potential, starting from the individual docking poses. The binding poses were selected using PMFs of dRMS between each snapshot of the gREST trajectory and the docking pose. dRMS is sensitive to the conformational/geometrical change of the ligand relative to the protein, serving as a good measure to classify different poses. In the calculation of dRMS, we treated the docking pose as the

reference structure (X^{ref}). The PMFs of dRMS for each FKBP–ligand complex are shown in Figure 3. In a 100 ns MD per replica, we could obtain the converged PMFs as a function of dRMS.

There are multiple basins in the PMFs. The PMFs for L2, L9, and L14 have several basins with similar stabilities. On the other hand, the single basin dominates in the other complexes, whereas there also exist multiple basins with much lower stabilities. There is no basin around dRMS = 0 Å, implying that

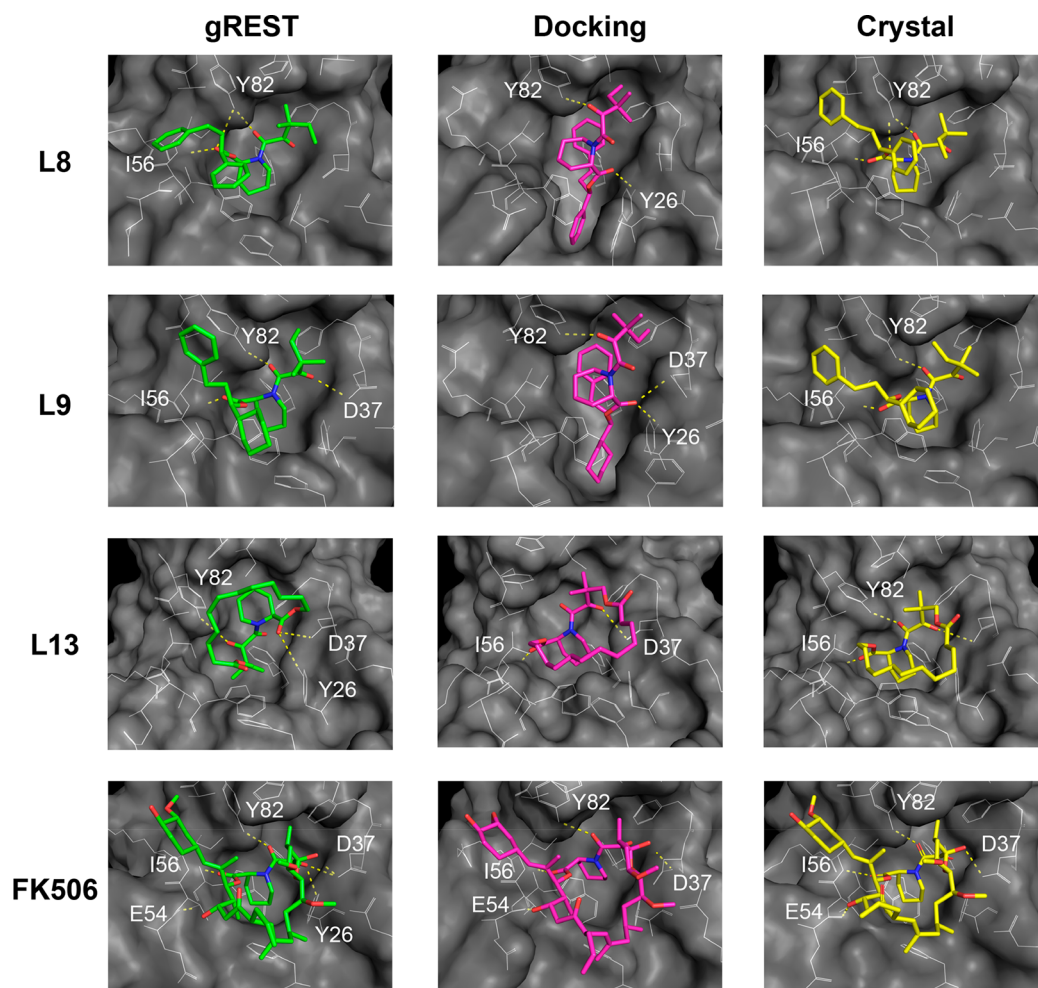


Figure 5. Conformations of the ligands (L8, L9, L13, and FK506) at the binding site. The ligands and surrounding residues of FKBP are shown in the stick and line representations, respectively. Hydrogen bonds are represented using dashed lines in yellow.

docking poses are not as stable in the flexible MD simulations. dRMSs of most ligands are larger than 2 Å. Exceptionally, the PMF for FK506, the largest ligand in the test set, has only one basin around 1 Å, and its binding pose is similar to the docking pose. We also calculated the PMFs of RMSD between the gREST trajectory and the docking pose (Figure S2). The number of basins in the PMFs of RMSD is lower than that of dRMS. For example, the dRMS PMFs for L2 and L12 have two or three basins, while the corresponding PMFs of RMSD have a single basin. This suggests that dRMS better classifies the binding poses compared to RMSD.

We obtained two binding poses for L2, L5, L9, L12, and L13, four binding poses for L14, and single binding pose for L3, L6, L8, and FK506. The binding poses were obtained from the basins within 1 kcal/mol of the lowest PMF(dRMS) as shown in Figure 4. In the binding poses of L2-2, L3, L8, L9-1, L12-1, and FK506, the pipecolate of the ligands faces the hydrophobic pocket of FKBP, consistent with the earlier study.¹⁷ On the other hand, the small ligands face other hydrophobic groups toward the hydrophobic pocket. For example, the cyclohexane is headed deep inside the pocket in L2-1, while the tert-pentyl group is inserted into the pocket in L5-1, L5-2, L6, and L9-2. For large ligands (L12, L13, and L14), there are multiple binding poses. In particular, L14 has four different binding poses, in which the ligand conformation changes one to the other. The tested ligands are flexible

because of the presence of a macrocycle and, thus, bind to the site with various poses.

To examine the protein–ligand interactions in more detail, the most stable poses of L8, L9, L13, and FK506 are compared with the corresponding docking poses and cocrystal structures (Figure 5). PDB IDs of cocrystal structures with L8, L9, L13, and FK506 are 1FKG, 1FKH, 1FKI, and 1FKJ, respectively. Hereafter the binding pose obtained by gREST and that of the X-ray crystal structure are referred to as “gREST pose” and “X-ray pose”, respectively. For L8, gREST successfully reproduces the X-ray pose (ligand’s RMSD of 2.1 Å). In both of the gREST and X-ray poses, L8 forms two hydrogen bonds with I56 and Y82, and the pipecolate of the ligand faces the hydrophobic pocket in the binding site. The docking pose of L8 also forms two hydrogen bonds, but two benzene rings face the binding site instead of the pipecolate (ligand’s RMSD of 6.2 Å). The binding pocket is expanded to accommodate these benzene rings, which may destabilize the docking pose. gREST also reproduces the X-ray pose of L9 (ligand’s RMSD of 2.2 Å). The protein–ligand interactions for L9 are similar to those of L8 except for one additional hydrogen bond with D37 in gREST pose. Similarly, the docking pose of L9 is almost the same as that of L8, although they are different from either gREST or X-ray poses (ligand’s RMSD of 5.8 Å). gREST and docking poses for FK506 are both in excellent agreement with the X-ray pose (ligand’s RMSDs of 1.2 and 1.9 Å for gREST

Table 1. Calculated Binding Affinity of Each Ligand with FKBP

ligand	$\Delta G_{\text{calc}}^{\text{gREST}^a}$ (kcal/mol)	$\Delta G_{\text{calc}}^{\text{docking}^b}$ (kcal/mol)	$\Delta G_{\text{calc}}^{\text{crystal}^c}$ (kcal/mol)	ΔG_{exp}^d (kcal/mol)
L2	-7.49 ± 0.35	-7.26 ± 0.56		-7.8
L3	-8.41 ± 0.21	-6.52 ± 0.33		-8.4
L5	-9.22 ± 0.32	-7.71 ± 0.59		-9.5
L6	-10.85 ± 0.41	-7.54 ± 0.72		-10.8
L8	-11.03 ± 0.34	-8.19 ± 0.46	-9.87 ± 0.41	-10.9
L9	-10.77 ± 0.37	-8.45 ± 0.64	-11.25 ± 0.31	-11.1
L12	-10.51 ± 0.31	-7.66 ± 0.62		-10.3
L13	-8.74 ± 0.27	-9.20 ± 0.50	-10.00 ± 0.23	-9.5
L14	-13.18 ± 0.47	-8.78 ± 0.53		-12.3
FK506	-15.81 ± 0.57	-15.32 ± 0.75	-15.39 ± 0.54	-12.8

^aCalculated binding affinities using the gREST poses. ^bCalculated binding affinities using the docking poses. ^cCalculated binding affinities using the X-ray poses. ^dBinding affinities obtained from the experiment.²⁸

and docking, respectively). The hydrogen bonds with D37, E54, I56, and Y82 stabilize the pose of FK506 in the binding site. The single basin of the PMF of dRMS (shown in Figure 3) means that FK506 does not move or rotate during the gREST simulation because of the hydrogen bond interactions.

Only for L13, gREST pose is markedly different from the X-ray pose (ligand's RMSD of 6.0 Å). gREST simulation for L13 was started from the docking pose, which is very close to the X-ray pose (ligand's RMSD of 1.4 Å). The ligand L13 was, however, turned over and faced its hydrophobic parts toward the pocket during the simulation to form the hydrogen bonds with Y26, D37, and Y82. These interactions seem to stabilize the gREST pose to the same extent as those in the crystal structure. The discrepancy between the gREST and the X-ray poses originates from the high flexibility of the macrocycle of L13. The effect of the flexibility of the macrocycle is discussed in section 4.

L9 and L13 have the second most stable poses (L9-2 and L13-2). Detailed interactions of these poses are shown in Figure S3. For L9-2, L9 inserts the tert-pentyl and cyclohexyl groups into the binding pocket, which stabilizes the hydrophobic interactions. However, L9 forms only one hydrogen bond with Y82. Similarly, L13-2 faces its hydrophobic parts toward the pocket like L13-1 and forms a hydrogen bond with the main chain of I56. For both ligands, the number of hydrogen bonds in the second most stable poses is smaller than that in the most stable poses.

The binding site residues slightly changes their conformation to accommodate the ligand. Figure S4 shows the ligand-induced conformational change from the apo structure to the global minimum pose bound to L8 (i.e., the gREST pose of L8). The side-chain atoms of Asp37, Phe46, Ile56, and Tyr82 move outward from the center of the binding pocket to create a space for the ligand, while His87 moves inward to interact with the ligand. The conformational change results in the tight packing between the ligand and the binding site atoms. We observed similar conformational changes for the other ligands too.

As with the dRMS analysis, we selected the binding pose corresponding to the global minimum of the PMF of RMSD. For L8, L9, L13, and FK506, we compared the pose obtained from the RMSD analysis with that obtained from the dRMS analysis (Figure S5). The binding poses of L8, L13, and FK506 perfectly match between the RMSD and dRMS analyses. In contrast, the binding pose of L9 from the RMSD analysis is different from that from the dRMS analysis and also deviates from the X-ray pose. This pose is similar to the pose

corresponding to the second local minimum of PMF(dRMS) (see Figure 4). This result implies that RMSD cannot differentiate two binding poses and mixes them in the lowest basin of its PMF.

3.3. Prediction of Binding Affinities Using FEP. We finally calculated the binding affinities of the gREST, docking, and X-ray poses using FEP and estimated correlation metrics (Pearson's r , root-mean-squared error (RMSE) and Kendall's τ) between the calculated affinities and the experimental values. In FEP calculations of the gREST poses, we assumed that the global minimum pose dominantly contributes to ΔG_{bind}^0 and used only the most stable poses (L2-1, L3, L5-1, L6, L8, L9-1, L12-1, L13-1, L14-1, and FK506 in Figure 4) for the binding affinity calculation. The calculated binding affinities of the docking poses deviate from the experimental values (Pearson's r of 0.68, RMSE of 2.40 kcal/mol, and Kendall's τ of 0.58) (Table 1 and Figure 6a). In particular, the deviations of L8, L9, L12, and L14 are large. In contrast, the calculated binding affinities of the gREST poses are in good agreement with the experimental values (Pearson's r of 0.95, RMSE of 1.04 kcal/mol, and Kendall's τ of 0.90) (Table 1 and Figure 6b). The calculated binding affinities for nine ligands are within 1 kcal/mol of error range from the experimental values. The calculated binding affinities of the X-ray poses fall into the same error range. The precision of FEP calculations critically depends on the reliability of binding pose.^{1,2} gREST successfully improves the binding pose obtained by Autodock Vina. Although the gREST pose of L13 is significantly different from the X-ray pose (Figure 5), its calculated binding affinity is similar to that of the X-ray pose. The gREST pose of L13 may exist as one of metastable states in solution, which is not observed by the X-ray crystallography and partially contributes to the affinity.

4. DISCUSSION

Figure 7 summarizes the flow of the protocol proposed in this study. First, the initial pose is built using any docking software. Second, the binding pose is refined using the gREST with the FB potential. Third, a few reliable poses are determined based on the PMF as a function of dRMS. Finally, the binding affinity of the most reliable pose is calculated using the FEP method. We referred to this protocol as the gREST+FEP method. In this study we employed only the FKBP-ligand complexes to benchmark our method. In our previous studies,^{26,27} we have successfully applied gREST to predict not only the X-ray structure of the complex of Src kinase and its inhibitor but also water positions around the binding site. In another example of

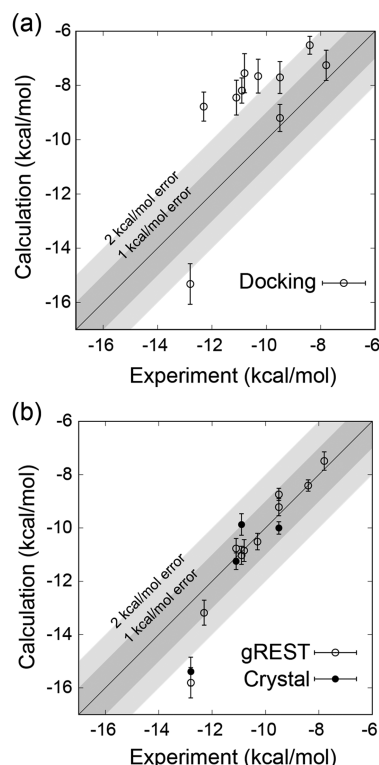


Figure 6. Calculated binding affinities for each FKBP-ligand complex. (a) Calculated binding affinities for the docking poses. (b) Calculated binding affinities using the gREST poses (white circle) and the X-ray poses (black circle). Experimental results are taken from ref 28.

the binding of T4 lysozyme and ligands, gREST also successfully reproduces cocrystal structures of three ligands by starting from an unbound state. Taken all together, gREST could be used for predicting the binding poses of various protein–ligand systems. Since the protocol proposed here does not necessarily require high-resolution cocrystal structures to predict binding affinities, gREST provides a promising tool to investigate the protein–ligand bindings lacking cocrystal structure information.

The present work illustrates how gREST combined with the FB restraint potential can predict the protein–ligand binding poses. This includes the refinement of the docking poses and the multiple binding poses, which are hardly determined by X-ray crystallography. Autodock Vina correctly predicted two out of four cocrystal structures (L13 and FK506), while gREST reproduced three out of four poses (L8, L9, and FK506). Although the gREST pose of L13 is significantly different from

the X-ray pose (Figure 5), its calculated binding affinity is similar to that of the X-ray pose. The gREST pose of L13 may exist as one of metastable states in solution, which is not observed by the X-ray crystallography, and partially contributes to the affinity. For the ligands lacking the cocrystal structures (L2, L3, L5, L6, L12, and L14), the calculated binding affinities of the predicted poses are in excellent agreement with the experimental observations, which also supports the reliability of the predicted poses. In principle, the gREST+FEP method can predict the binding pose and affinity regardless of the initial structure used, although Autodock Vina was used in this study. In fact, for L8, gREST has reproduced the X-ray pose from the docking pose that is significantly different from the crystal structure (see Figure 5).

For L13, the calculated binding affinities from FEP calculations indicate that the docking and X-ray poses are more stable than the gREST pose (see Table 1). This inconsistency between PMF(dRMS) and the calculated binding affinity can be caused by high flexibility of the ligand. Figure 8a shows the conformation of L13 in the gREST pose and that in the X-ray pose, which are aligned on the pipicolates. In the L13 conformation of the gREST pose two rotational bonds (dotted circles in Figure 8a) have different orientation from those in the X-ray pose, inducing the large deformation of the macrocycle (Time series of the conformation change is shown in Figure S6). Due to the deformation the pipicolate of L13 cannot face the hydrophobic pocket. The flexibility of L13's macrocycle is also suggested by the experiment (L13 has two bound conformations, whose macrocycles are slightly different. In addition, the unbound conformation of L13 is different from those at the bound state).²⁸ The interconversion of macrocycle conformations might not occur within the 100 ns gREST simulation, preventing the sampling of X-ray pose. Two rotational bonds of L13 may become too flexible due to the inaccuracy of the corresponding dihedral parameters or high solute temperatures in gREST simulations. To suppress the ligand flexibility, we performed two gREST simulations with the conformational restraint. First simulation was started from the best docking pose, and the conformation of L13 was restrained to the initial conformation by applying the RMSD restraint (force constant of 10 kcal/mol/Å²). The setups and analyses for the simulation were the same as used in section 3.2. The PMF(dRMS) has only one basin around 1 Å, and the corresponding pose is almost the same as the X-ray pose (Figure 8b). Second simulation was started from another docking pose (the 10th highest affinity in Autodock Vina) with the same restraint. Even though the initial pose was totally different from the X-ray pose, the PMF(dRMS) of L13

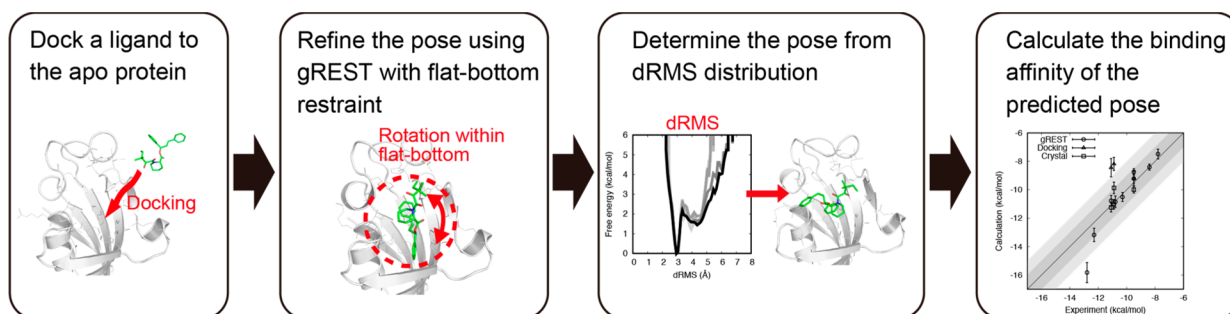


Figure 7. Flow of the gREST+FEP method for binding affinity prediction.

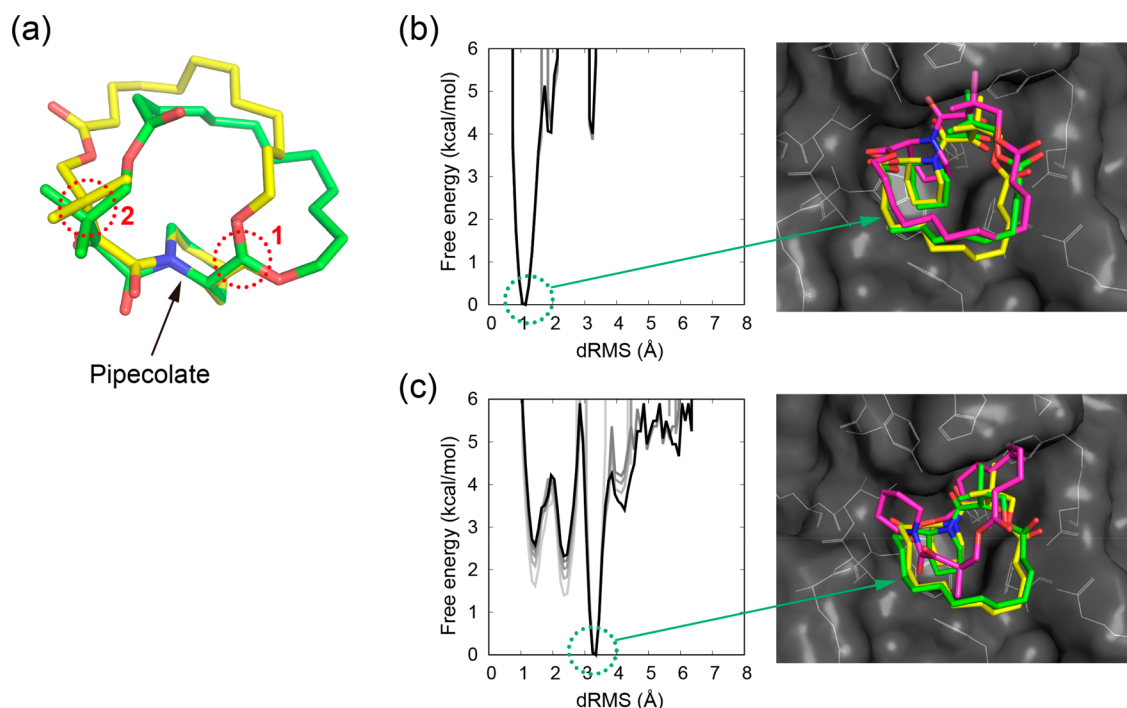


Figure 8. (a) Conformations of L13 at the binding site. The conformation determined by the gREST simulation and that in the X-ray crystal structure are shown in green and yellow, respectively. The dotted circles indicate the large differences between gREST and X-ray conformations. (b, c) PMF(*dRMS*) and conformation of L13 from gREST simulations with the conformational restraint applied to the ligand. The gREST pose, docking pose, and X-ray pose are shown in green, magenta, and yellow, respectively. The gREST simulations were started from the docking pose with the highest docking score (b) and that with the 10th highest score (c).

converged, and its most stable basin corresponds to the X-ray pose (Figure 8c). For ligands having large conformational flexibility, such as a compound with macrocycle, one may apply the RMSD restraint to suppress the ligand flexibility, improving the convergence of gREST simulations.

The protein residues in the binding site as well as the ligand are selected as the solute region in the gREST simulations. The flexibility of these residues is enhanced (Figure 2e), accelerating the reorganization of the protein–ligand interaction in response to the change in the binding pose. The enhancement of the protein flexibility is an advantage over the other methods in which only the protein–ligand interactions are scaled to search the best binding pose.^{2,19,20} Our method is similar to FEP+⁴⁶ or FEP/REST,^{47,48} in which the sampling efficiency in predefined solute region is enhanced using REST2.²⁵ However, gREST provides a more flexible framework for solute definition than REST2. gREST allows to select the part of energy terms, such as dihedral angle and CMAP⁴⁹ energy terms, of the ligand and protein as the solute region. The proper solute definition could further enhance the sampling of protein–ligand poses with a smaller number of replicas. gREST also weakens interactions between the binding site residues and water molecules at high solute temperatures, and thereby water diffusion near the binding site is accelerated at the temperatures, which enable to reproduce water positions in the X-ray structure.²⁶ A possible future application of gREST includes the prediction of affinity changes due to protein mutations relevant for drug resistance.^{50–54} gREST, with enhanced flexibility of the residues around the mutated site, could predict both the changes in the protein conformation and ligand pose upon a mutation.

In this study, we selected HIE for all histidine residues, because their tautomer states seem not to largely affect

interactions between the residues and the ligand. However, in general, accurate determination of protonation and tautomer states of histidine residues are very important on predictions of the pose and affinity. The proton position or protonation state significantly affects protein–ligand interactions as well as water-mediated interactions. Constant pH MD simulations allows all three histidine states (HIE/HID/HIP) to be sampled and improve the binding pose and affinity predictions.^{55,56} Constant pH MD simulations will be required for the predictions with higher accuracy in future drug discovery.

The present gREST simulations used only 12 replicas to enhance the ligand rotational motions in the binding site sufficiently for the pose prediction. However, the computational cost for simulating tens of replicas is still large to apply gREST for virtual screening of a large number of drug candidates (see Table S1). The gREST method may be replaced with a single-replica enhanced sampling methods, such as the simulated tempering^{57–59} or Gaussian accelerated MD,^{60,61} to reduce the number of replicas. Those methods should be modified to accelerate the movement or flexibility of only a ligand and relevant residues of a target protein like gREST. Further developments of the method along the line would improve the efficiency and precision of the future screening of drug candidates.

To calculate the absolute binding affinities using the double annihilation method, we have implemented the alchemical FEP method into GENESIS.³¹ In GENESIS, the alchemical simulations at different coupling parameters λ^{el} and λ^{lj} are performed in parallel, and Hamiltonian replica exchange can be used by exchanging the parameters. The double decoupling method^{15,16,44} are also available by adding restraints between a ligand and a target protein. In addition to the absolute binding affinity, GENESIS also supports the alchemical calculation of

relative binding affinities with both dual-topology^{62,63} and hybrid-topology⁶⁴ frameworks. After predicting the binding pose of a ligand using gREST, the relative binding affinity from the ligand to another one can be calculated using GENESIS. The FEP-implemented version of GENESIS is freely available, aiming to contribute the acceleration of in-silico drug discovery for a wide range of diseases.

5. CONCLUSION

We have proposed a new protocol, referred to as the gREST +FEP, to predict protein–ligand binding poses and affinities without prior knowledge of X-ray crystal structures. We have tested the method on the well-established benchmark system, the binding of ten ligands to FKBP. gREST enhances the reorientation/rotation of the ligands and the flexibility of protein residues in the binding site, resulting in the precise prediction of the multiple binding poses including the X-ray structures. The calculated binding affinities of the predicted poses, which are calculated using FEP, excellently reproduce the experimental values. The gREST+FEP method might be generally applicable to a broad range of protein–ligand bindings and particularly useful when high-resolution cocrystal structures are unavailable.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.0c00338>.

Computational cost for the gREST+FEP method (Table S1), time series of free energies of bound and unbound states in FEP calculations (Figure S1), PMFs of RMSD for each FKBP–ligand complex at 300 K (Figure S2), conformations of the second most stable poses (L9-2 and L13-2) at the binding site (Figure S3), conformational change upon L8 binding (Figure S4), conformations of the ligands (L8, L9, L13, and FK506) at the binding site (Figure S5), and time series of the conformational RMSD of L13 (Figure S6) (PDF)

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Notes

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